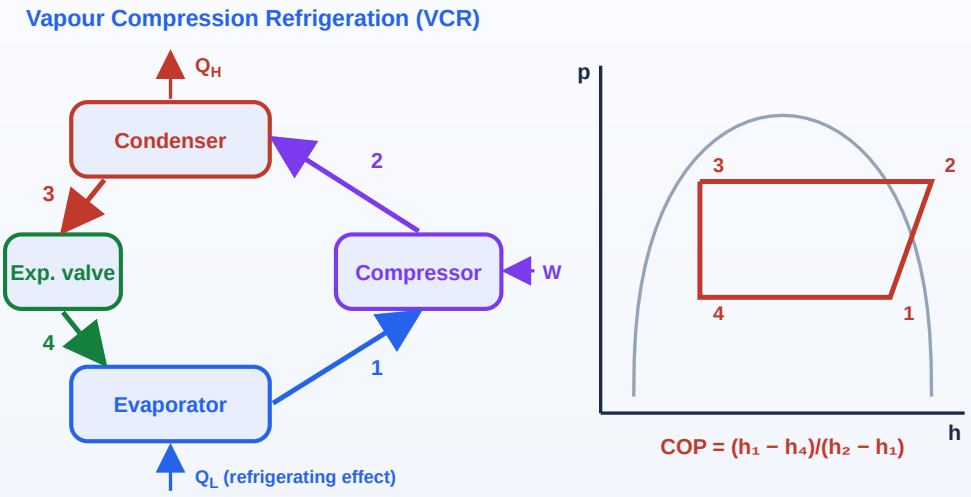
UPSSSC ALERT

UPSSSC JE

MECHANICAL REFRIGERATION & AIR COMPRESSION

Complete English Edition

Unit 7 · 5 Marks · 2026



UPSSSC JE (Mechanical) 2026: Part-1 has **5 questions (5 marks)** from **Refrigeration and Air Compression**. Syllabus: air compressors and their cycles; principles of refrigeration plants.

Each chapter contains key points, diagrams, formula table, detailed explanation, solved numericals, 30 one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (C. P. Arora, R. S. Khurmi, R. K. Rajput).

For a better score, study this PDF and also attempt the QUIZ.

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Chapter 1: Air Compressors & their Cycles

Air Compressors & their Cycles – Basic Concept

An **air compressor** takes in atmospheric air and raises its pressure. Compressed air is used for pneumatic tools (drills, hammers), spray painting, tyre inflation, brakes (trains/buses), gas turbines, supercharging and refrigeration. Work must be supplied to drive a compressor.

1. Key Points

- **Classification:** reciprocating (piston–cylinder; low flow, high pressure) and rotary (Roots blower, vane, screw, centrifugal, axial; high flow, lower pressure ratio).
- **Types of compression:** isothermal ($pV = C$) — minimum work; polytropic ($pV^n = C$, $1 < n < \gamma$) — practical; adiabatic ($pV^\gamma = C$) — maximum work.
- **Isothermal work:** $W = p_1V_1 \ln(p_2/p_1)$. **Polytropic work:** $W = [n/(n-1)]\, p_1V_1 [(p_2/p_1)^{(n-1)/n} - 1]$.
- **Isothermal efficiency:** η_{iso} = isothermal work ÷ actual work. Cooling (water jacket, fins) is needed to reduce work.
- **Clearance volume:** compressed air left at TDC re-expands before fresh air enters — this lowers volumetric efficiency but does not change work per kg.
- **Volumetric efficiency:** $\eta_v = 1 + C - C(p_2/p_1)^{1/n}$, where $C = V_c/V_s$. η_v falls as pressure ratio rises.
- **Multi-stage compression:** two or more stages with an intercooler in between for high pressure; reduces work, improves volumetric efficiency and approaches isothermal compression.
- **Perfect intercooling and minimum work:** for two stages the intermediate pressure $p_2 = \sqrt{(p_1p_3)}$; then work in both stages is equal. For N stages the pressure ratio per stage = $(p_{N+1}/p_1)^{1/N}$.
- **Free air delivered (FAD):** volume flow of compressed air reduced to atmospheric conditions (usually 1.013 bar, 15°C).
- **Centrifugal compressor:** the impeller accelerates air radially and the diffuser converts velocity into pressure; used in gas turbines, jet engines and superchargers.

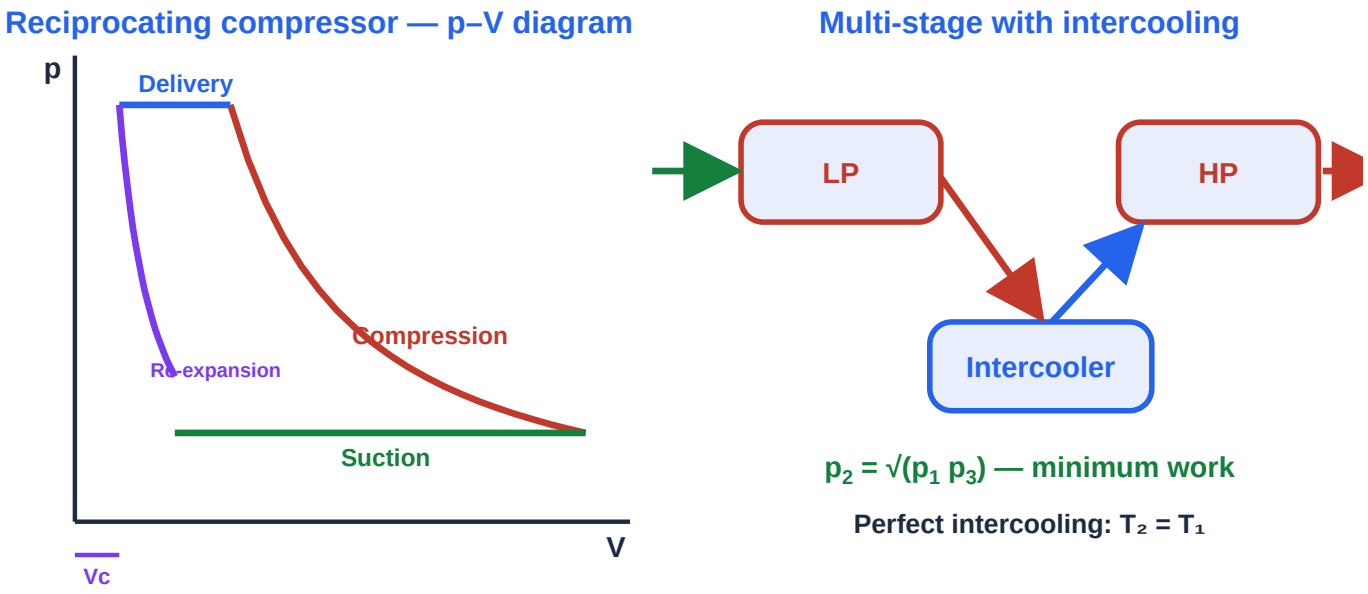


Fig 7.1: Reciprocating air compressor and two-stage compression

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Isothermal work	$W = p_1V_1 \ln(p_2/p_1)$
Polytropic work	$W = [n/(n-1)]\, p_1V_1\, [(p_2/p_1)^{(n-1)/n} - 1]$
Adiabatic work	$W = [\gamma/(\gamma-1)]\, p_1V_1\, [(p_2/p_1)^{(\gamma-1)/\gamma} - 1]$
Isothermal efficiency	$\eta_{iso} = W_{iso}/W_{actual}$
Volumetric efficiency	$\eta_v = 1 + C - C(p_2/p_1)^{1/n}$
Two-stage intermediate pressure	$p_2 = \sqrt{(p_1\, p_3)}$
N stages — ratio per stage	$(p_{final}/p_1)^{1/N}$
N-stage minimum work	$W = N[n/(n-1)]p_1V_1[(p_f/p_1)^{(n-1)/(Nn)} - 1]$
Delivery temperature	$T_2 = T_1\, (p_2/p_1)^{(n-1)/n}$
Indicated power	$P = W \times N/60$ (single acting)

3. Detailed Explanation

- **Why isothermal is ideal:** on the p–V diagram the isothermal curve is the least steep, so the area (work) is minimum; in adiabatic compression the rise in temperature keeps volume larger, needing more work.
- **Benefit of intercooler:** cooling air to the initial temperature after the first stage reduces volume entering the second stage, saving work; it also prevents lubricating oil from burning at high temperature.
- **Single and double acting:** single acting compresses on one side of the piston; double acting on both sides — two deliveries per revolution.

- **Rotary compressors:** the Roots blower has no internal compression (back-flow); the vane compressor has internal compression; axial compressors handle the largest flows.
- **Effect of clearance on indicator diagram:** larger clearance reduces the volume drawn in, but work per kg is unchanged because the re-expanding clearance air returns its work.

4. Comparison Table — Reciprocating vs Rotary Compressor

Basis	Reciprocating	Rotary
Flow	Low	High
Pressure ratio	High (multi-stage)	Low per stage
Speed	Low	High
Delivery	Pulsating	Continuous, smooth
Lubrication & air purity	Oil may mix	Clean air

4. Comparison Table — Isothermal, Polytropic and Adiabatic Compression

Basis	Isothermal	Polytropic	Adiabatic
Law	$pV = C$	$pV^n = C$	$pV^\gamma = C$
Work	Minimum	Intermediate	Maximum
Delivery temperature	Equal to T_1	Moderate	Highest

5. Solved Numerical Problems

Example 1. 1 m³ of air is compressed isothermally from 1 bar to 7 bar. Find the work.

Solution:

1. $W = 100 \times 1 \times \ln 7 = 100 \times 1.946 =$ **194.6 kJ**

Example 2. Solve the above for polytropic compression with n = 1.3 and find the isothermal efficiency.

Solution:

1. $(7)^{0.3/1.3} = 7^{0.2308} = 1.567$
2. $W = (1.3/0.3) \times 100 \times (1.567 - 1) = 4.333 \times 56.7 =$ **245.7 kJ**
3. $\eta_{iso} = 194.6/245.7 =$ **79.2%**

Example 3. A two-stage compressor compresses from 1 bar to 16 bar. Find the intermediate pressure for minimum work.

Solution:

1. $p_2 = \sqrt{(1 \times 16)} =$ **4 bar**

Example 4. Clearance ratio C = 0.05, pressure ratio 8, n = 1.3. Find the volumetric efficiency.

Solution:

1. $8^{1/1.3} = 8^{0.769} = 4.95$
2. $\eta_v = 1 + 0.05 - 0.05 \times 4.95 =$ **80.2%**

Example 5. Air at 27°C is compressed from 1 to 8 bar with n = 1.3. Find the delivery temperature.

Solution:

1. $T_2 = 300 \times 8^{0.3/1.3} = 300 \times 1.616 =$ **484.8 K (211.8°C)**

6. Frequently Asked Facts

- Isothermal compression needs minimum work, adiabatic the maximum.
- Two-stage minimum work: $p_2 = \sqrt{(p_1 p_3)}$, equal work in both stages.
- Clearance lowers volumetric efficiency, not work per kg.
- Isothermal efficiency = isothermal work ÷ actual work.

7. Note (Confusing Facts)

- Volumetric efficiency decreases (not increases) as pressure ratio rises.
- There is no internal compression in a Roots blower.
- Perfect intercooling means the second-stage inlet temperature equals the first-stage inlet temperature.

Remember: One line: the lower n, the lower the work (n = 1 isothermal). Assertion–Reason: (A) Multi-stage compression is used for high pressure. (R) Intercooling reduces work and improves volumetric efficiency. — Both true; R explains A.

30 Most Important One-Liner Facts — Air Compressors & their Cycles

1. Compression needing least work?

— Isothermal.

2. Compression needing most work?

— Adiabatic.

3. Practical compression?

— Polytropic ($pV^n = C$).

4. Isothermal work?

— $p_1V_1 \ln(p_2/p_1)$.

5. Isothermal efficiency?

— Isothermal ÷ actual work.

6. Volumetric efficiency formula?

— $1 + C - C(p_2/p_1)^{1/n}$.

7. Clearance ratio?

— V_c/V_s.

8. η_v with more clearance?

— Decreases.

9. η_v with higher pressure ratio?

— Decreases.

10. Clearance effect on work per kg?

— None.

11. Device between stages?

— Intercooler.

12. Two-stage intermediate pressure?

— $\sqrt{(p_1 p_3)}$.

13. Stage works at minimum?

— Equal.

14. Ratio per stage for N stages?

— $(p_f/p_1)^{1/N}$.

15. Perfect intercooling?

— $T_2 = T_1$.

16. Benefit of multi-staging?

— Less work, higher η_v.

17. FAD stands for?

— Free Air Delivered.

18. No internal compression?

— Roots blower.

19. Compressor in jet engine?

— Axial / centrifugal.

20. Pressure-raising part of centrifugal?

— Diffuser.

21. Low flow, high pressure?

— Reciprocating.

22. High flow, low pressure?

— Rotary.

23. Means of compressor cooling?

— Water jacket, fins.

24. Compression on both sides?

— Double acting.

25. Delivery temperature?

— $T_1(p_2/p_1)^{(n-1)/n}$.

26. Extra benefit of intercooler?

— Lube oil does not burn.

27. Clearance air process on p–V?

— Re-expansion.

28. Use of compressed air?

— Pneumatic tools, brakes.

29. 1 to 16 bar two-stage — p_2?

— 4 bar.

30. γ of air?

— 1.4.

Practice MCQs — Air Compressors & their Cycles

Q1. Minimum work in air compression is for:

(a) Adiabatic
(b) Isothermal
(c) Polytropic
(d) Isentropic

✔ **Answer: (b) Isothermal**

Explanation: pV = C.

Q2. With increased clearance volume, volumetric efficiency:

(a) Increases
(b) Decreases
(c) Constant
(d) Zero

✔ **Answer: (b) Decreases**

Explanation: Re-expansion reduces intake.

Q3. Intermediate pressure for minimum work in a two-stage compressor:

(a) $(p_1 + p_3)/2$
(b) $\sqrt{(p_1p_3)}$
(c) p_1p_3
(d) $p_3 - p_1$

✔ **Answer: (b) $\sqrt{(p_1p_3)}$**

Explanation: Geometric mean.

Q4. Isothermal efficiency =

(a) Actual ÷ isothermal work
(b) Isothermal ÷ actual work
(c) Adiabatic ÷ isothermal
(d) FAD ÷ swept

✔ **Answer: (b) Isothermal ÷ actual work**

Explanation: Ideal ÷ actual.

Q5. Two-stage from 1 to 9 bar — intermediate pressure:

(a) 3 bar
(b) 4.5 bar
(c) 5 bar
(d) 9 bar

✔ **Answer: (a) 3 bar**

Explanation: √9.

Q6. Purpose of intercooling:

(a) Increase work
(b) Reduce work
(c) Reduce pressure
(d) Reduce flow

✔ **Answer: (b) Reduce work**

Explanation: Approaches isothermal.

Q7. In a Roots blower:

(a) Internal compression occurs
(b) No internal compression
(c) There is a piston
(d) There are valves

✔ **Answer: (b) No internal compression**

Explanation: Pressure rises by back-flow.

Q8. Volumetric efficiency as pressure ratio increases:

(a) Increases
(b) Decreases
(c) Constant
(d) First rises then falls

✔ **Answer: (b) Decreases**

Explanation: $(p_2/p_1)^{1/n}$ increases.

Q9. Effect of clearance on work per kg:

(a) Increases
(b) Decreases
(c) No effect
(d) Doubles

✔ **Answer: (c) No effect**

Explanation: Re-expansion returns work.

Q10. 1 m^3 air from 1 to e (2.718) bar isothermally — work:

(a) 100 kJ
(b) 271.8 kJ
(c) 50 kJ
(d) 1 kJ

✔ **Answer: (a) 100 kJ**

Explanation: 100 × ln e = 100.

Q11. Perfect intercooling means:

(a) Second-stage inlet temp = initial temp
(b) Equal pressure
(c) No cooling
(d) Adiabatic compression

✔ **Answer: (a) Second-stage inlet temp = initial temp**

Explanation: $T_2 = T_1$.

Q12. Compressor suited to very high flow:

(a) Reciprocating
(b) Axial
(c) Vane
(d) Roots

✔ **Answer: (b) Axial**

Explanation: Axial.

Q13. Pressure ratio per stage in an N-stage compressor:

(a) $p_f/(Np_1)$
(b) $(p_f/p_1)^{1/N}$
(c) $N(p_f/p_1)$
(d) $(p_f/p_1)^N$

✔ **Answer: (b) $(p_f/p_1)^{1/N}$**

Explanation: Equal ratios.

Q14. In a centrifugal compressor velocity is converted to pressure in the:

(a) Impeller
(b) Diffuser
(c) Shaft
(d) Inlet

✔ **Answer: (b) Diffuser**

Explanation: Diffuser.

Q15. C = 0.05, $p_2/p_1 = 8$, n = 1.3 — η_v about:

(a) 70%
(b) 80%
(c) 90%
(d) 95%

✔ **Answer: (b) 80%**

Explanation: $1.05 - 0.05 \times 4.95$.

SSC JE / UPSSSC JE Previous Year Questions — Air Compressors & their Cycles

Q1. Isothermal efficiency of an air compressor is highest when compression is: SSC JE (Previous Year)

(a) Adiabatic
(b) Close to isothermal
(c) Isentropic
(d) Constant volume

✔ **Answer: (b) Close to isothermal**

Explanation: Good cooling.

Q2. Advantage of multi-stage compression: SSC JE (Previous Year)

(a) Reduction in work
(b) Reduction in volumetric efficiency
(c) Higher temperature
(d) More work

✔ **Answer: (a) Reduction in work**

Explanation: Intercooling.

Q3. Clearance volume in a reciprocating compressor reduces its: SSC JE (Previous Year)

(a) Volumetric efficiency
(b) Isothermal efficiency
(c) Mechanical efficiency
(d) Pressure

✔ **Answer: (a) Volumetric efficiency**

Explanation: Less intake volume.

Q4. For minimum work in a two-stage compressor: SSC JE (Previous Year)

(a) Equal pressure ratio in both stages
(b) Higher in first stage
(c) Higher in second stage
(d) No intercooler

✔ **Answer: (a) Equal pressure ratio in both stages**

Explanation: $p_2/p_1 = p_3/p_2$.

Q5. Purpose of cooling in a compressor: SSC JE (Previous Year)

(a) Reduce work
(b) Increase work
(c) Reduce pressure
(d) Increase flow

✔ **Answer: (a) Reduce work**

Explanation: Towards isothermal.

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Chapter 2: Principles of Refrigeration Plants

Principles of Refrigeration Plants – Basic Concept

Refrigeration is the process of removing heat from a space or substance to bring and keep its temperature below the surroundings. By the second law (Clausius) this is impossible without external work. Domestic refrigerators, air conditioners, cold storages, ice plants and water coolers are examples.

1. Key Points

- **Unit of refrigeration:** 1 ton of refrigeration (TR) = rate of freezing 1 ton (907 kg) of water at 0°C into ice in 24 hours = 211 kJ/min ≈ 3.5 kW (often taken as 210 kJ/min).
- **COP:** COP = refrigerating effect ÷ work input = Q_L/W; it can exceed 1. Carnot COP = $T_L/(T_H - T_L)$.
- **Vapour compression refrigeration (VCR):** four components — compressor (low-pressure vapour → high-pressure superheated vapour), condenser (heat rejected, vapour → liquid), expansion valve (throttling, h constant, pressure and temperature fall), evaporator (heat absorbed, liquid → vapour).
- **COP of VCR (from p–h chart):** COP = $(h_1 - h_4)/(h_2 - h_1)$, where $h_4 = h_3$ (throttling).
- **Vapour absorption refrigeration (VAR):** an absorber–generator replaces the compressor; energy is supplied as heat (steam, solar, waste heat). NH_3–H_2O (ammonia refrigerant, water absorbent) and LiBr–H_2O (water refrigerant) systems. The Electrolux refrigerator also uses hydrogen.
- **Air refrigeration (Bell–Coleman / reversed Brayton):** working fluid is air; used for aircraft cabin cooling; low COP.
- **Refrigerants:** R-12 (Freon-12, CFC – ozone depleting, banned), R-22 (HCFC), R-134a (HFC – domestic fridges/car AC), R-717 (ammonia – large cold stores, ice plants; toxic), R-744 (CO_2), R-600a (isobutane), R-290 (propane), R-718 (water).
- **Properties of a good refrigerant:** high latent heat, low boiling point, moderate condensing pressure, non-toxic, non-flammable, ozone-friendly (ODP = 0) and low GWP, cheap, easy leak detection.
- **Subcooling and superheating:** subcooling the liquid after the condenser and superheating vapour after the evaporator both increase refrigerating effect; subcooling raises COP.
- **Link to air conditioning:** psychrometry – dry bulb, wet bulb, dew point, relative humidity; comfort around 25°C DBT, 50% RH.

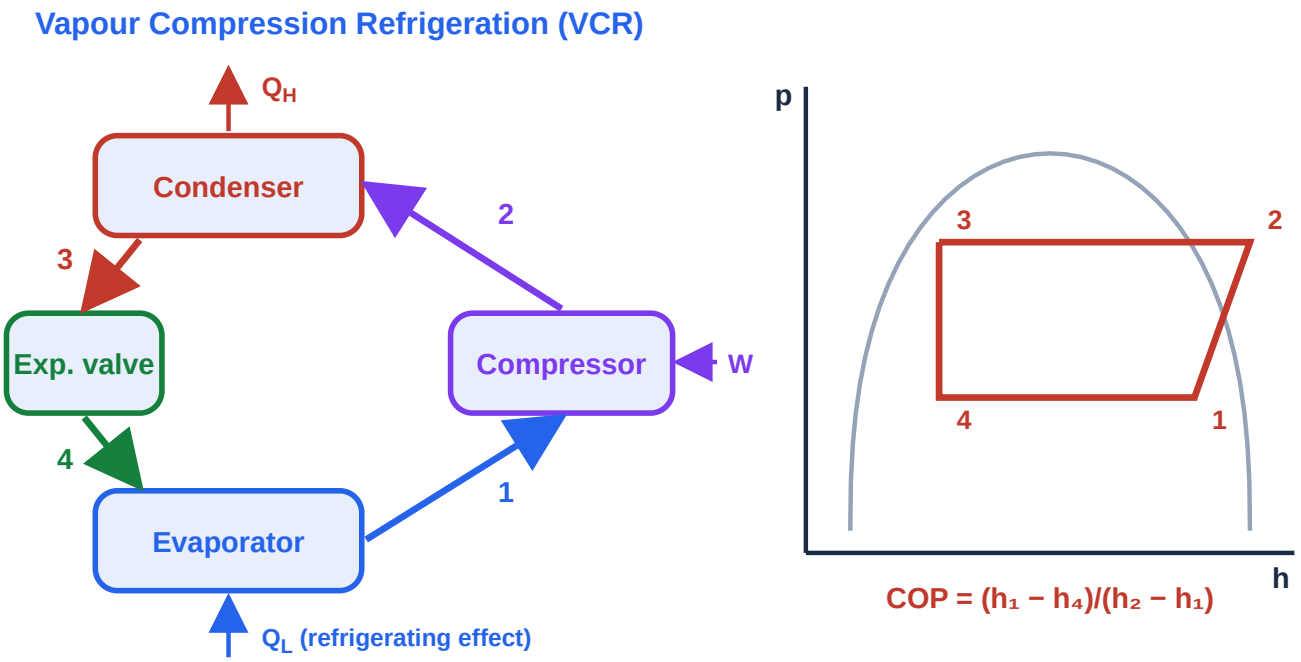


Fig 7.2: VCR cycle — components and p–h diagram

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
COP (refrigerator)	**$COP = Q_L/W$**
Carnot COP	**$T_L/(T_H - T_L)$**
VCR COP	**$(h_1 - h_4)/(h_2 - h_1)$**
Refrigerating effect	**$R_E = h_1 - h_4$**
Compressor work	**$W = h_2 - h_1$**
Condenser heat	**$Q_H = h_2 - h_3$**
Throttling	**$h_3 = h_4$**
Refrigerant mass flow	**$m = 211 \times TR/(h_1 - h_4)$ kg/min**
1 TR	**211 kJ/min = 3.517 kW**
HP COP	**$COP_{HP} = COP_R + 1$**

3. Detailed Explanation

- **Working of VCR:** refrigerant boils at low pressure in the evaporator absorbing heat → compressor raises pressure and temperature → condenser rejects heat to air/water and liquefies it → expansion valve drops pressure and the cold mixture returns to the evaporator.
- **VCR vs VAR:** VCR has high COP, is electrically driven and compact; VAR has low COP, is heat driven, silent (few moving parts) and bulky — used where waste heat is available.
- **Domestic refrigerator:** hermetically sealed compressor, capillary tube expansion device, air-cooled condenser (rear grille), R-134a/R-600a.

- **Expansion devices:** capillary tube (small units), thermostatic expansion valve (TXV – AC), automatic expansion valve, float valve (large plants).
- **Improving COP:** raise evaporator temperature, lower condenser temperature, subcool; the smaller the temperature lift, the higher the COP.

4. Comparison Table — Vapour Compression vs Vapour Absorption

Basis	VCR	VAR
Energy input	Mechanical work (electricity)	Heat (steam, solar, waste)
Key device	Compressor	Absorber + generator
COP	High (3–5)	Low (≈ 0.5–1)
Noise & moving parts	More	Very few (silent)
Refrigerant	R-134a, R-22 etc.	NH_3 (water absorbent) / water (LiBr)

5. Solved Numerical Problems

Example 1. A Carnot refrigerator works between −10°C and 30°C. Find its COP.

Solution:

1. T_L = 263 K, T_H = 303 K
2. COP = 263/40 = **6.58**

Example 2. In a VCR cycle h_1 = 400, h_2 = 440, $h_3 = h_4$ = 250 kJ/kg. Find refrigerating effect, work and COP.

Solution:

1. R_E = 400 − 250 = **150 kJ/kg**
2. W = 440 − 400 = **40 kJ/kg**
3. COP = 150/40 = **3.75**

Example 3. The above plant has 5 TR capacity. Find refrigerant mass flow and compressor power.

Solution:

1. m = 211 × 5/150 = **7.03 kg/min**
2. P = 7.03 × 40/60 = **4.69 kW**

Example 4. An AC removes 3.5 kW and consumes 1 kW. Find COP and capacity in TR.

Solution:

1. COP = 3.5/1 = **3.5**
2. Capacity ≈ 3.5/3.517 ≈ **1 TR**

Example 5. Find the COP if the above Carnot refrigerator is run as a heat pump.

Solution:

1. COP_{HP} = 6.58 + 1 = **7.58**

6. Frequently Asked Facts

- 1 TR = 211 kJ/min ≈ 3.5 kW.
- Enthalpy is constant across the expansion valve in VCR ($h_3 = h_4$).
- NH_3 – ice plants/cold storage; R-134a – domestic fridges; in LiBr–H_2O water is the refrigerant.
- Air refrigeration (Bell–Coleman) – aircraft.

7. Note (Confusing Facts)

- In NH_3–H_2O ammonia is the refrigerant and water the absorbent; in LiBr–H_2O it is reversed — water is the refrigerant.
- A COP above 1 does not violate the second law.
- Heat leaves at the condenser and enters at the evaporator.

Remember: One line: compressor → condenser → expansion valve → evaporator. Assertion–Reason: (A) R-12 has been phased out. (R) It depletes the ozone layer. — Both true; R explains A.

30 Most Important One-Liner Facts — Principles of Refrigeration Plants

1. 1 TR = ?

— 211 kJ/min ≈ 3.5 kW.

2. COP?

— Refrigerating effect ÷ work.

3. Carnot refrigerator COP?

— $T_L/(T_H - T_L)$.

4. Four components of VCR?

— Compressor, condenser, expansion valve, evaporator.

5. Where is heat absorbed?

— Evaporator.

6. Where is heat rejected?

— Condenser.

7. Process in expansion valve?

— Throttling (h constant).

8. VCR COP formula?

— $(h_1 - h_4)/(h_2 - h_1)$.

9. Refrigerating effect?

— $h_1 - h_4$.

10. Energy input in VCR?

— Mechanical work.

11. Energy input in VAR?

— Heat.

12. Replaces compressor in VAR?

— Absorber and generator.

13. Refrigerant in NH_3–H_2O?

— Ammonia.

14. Refrigerant in LiBr–H_2O?

— Water.

15. Third gas in Electrolux?

— Hydrogen.

16. Aircraft cabin cooling?

— Air refrigeration (Bell–Coleman).

17. R-717?

— Ammonia.

18. R-744?

— CO_2.

19. R-718?

— Water.

20. Domestic fridge refrigerant?

— R-134a / R-600a.

21. Ozone-depleting refrigerant?

— R-12 (CFC).

22. Ice plant refrigerant?

— Ammonia.

23. Expansion device in fridge?

— Capillary tube.

24. Expansion device in AC?

— Thermostatic expansion valve.

25. Effect of subcooling?

— COP increases.

26. Feature of VAR?

— Silent, heat driven.

27. Refrigerant state after compressor?

— High-pressure superheated vapour.

28. COP with higher evaporator temp?

— Increases.

29. Comfort DBT and RH?

— ≈ 25°C, 50%.

30. Refrigeration is based on?

— Second law (Clausius).

Practice MCQs — Principles of Refrigeration Plants

Q1. 1 ton of refrigeration equals:

(a) 21 kJ/min
(b) 211 kJ/min
(c) 2110 kJ/min
(d) 3.5 kJ/min

✔ **Answer: (b) 211 kJ/min**

Explanation: ≈ 3.5 kW.

Q2. In VCR the refrigerant absorbs heat in the:

(a) Condenser
(b) Evaporator
(c) Compressor
(d) Expansion valve

✔ **Answer: (b) Evaporator**

Explanation: Evaporator.

Q3. Process in the expansion valve:

(a) Isentropic
(b) Isenthalpic (throttling)
(c) Isothermal
(d) Isobaric

✔ **Answer: (b) Isenthalpic (throttling)**

Explanation: h constant.

Q4. In VAR the compressor is replaced by:

(a) Pump
(b) Absorber and generator
(c) Turbine
(d) Nozzle

✔ **Answer: (b) Absorber and generator**

Explanation: Thermal compression.

Q5. Refrigerant in a LiBr–water system:

(a) LiBr
(b) Water
(c) Ammonia
(d) Hydrogen

✔ **Answer: (b) Water**

Explanation: LiBr is the absorbent.

Q6. Used for aircraft air conditioning:

(a) VCR
(b) VAR
(c) Air refrigeration
(d) Steam jet

✔ **Answer: (c) Air refrigeration**

Explanation: Light, air available.

Q7. $h_1 = 400$, $h_2 = 440$, $h_4 = 250$ — COP:

(a) 3.75
(b) 4.75
(c) 2.5
(d) 6

✔ **Answer: (a) 3.75**

Explanation: 150/40.

Q8. Carnot refrigerator COP between −10°C and 30°C:

(a) 0.33
(b) 6.58
(c) 7.58
(d) 10

✔ **Answer: (b) 6.58**

Explanation: 263/40.

Q9. R-717 is:

(a) Freon
(b) Ammonia
(c) CO_2
(d) Water

✔ **Answer: (b) Ammonia**

Explanation: Ammonia.

Q10. Expansion device in a domestic refrigerator:

(a) Capillary tube
(b) Float valve
(c) TXV
(d) Nozzle

✔ **Answer: (a) Capillary tube**

Explanation: Simple, cheap.

Q11. Most damage to the ozone layer is by:

(a) R-134a
(b) R-12
(c) R-717
(d) R-744

✔ **Answer: (b) R-12**

Explanation: CFC.

Q12. Which is not a desirable refrigerant property?

(a) High latent heat
(b) Toxicity
(c) Non-flammable
(d) Low boiling point

✔ **Answer: (b) Toxicity**

Explanation: Should be non-toxic.

Q13. State of refrigerant after the condenser in VCR:

(a) High-pressure liquid
(b) Low-pressure vapour
(c) High-pressure vapour
(d) Solid

✔ **Answer: (a) High-pressure liquid**

Explanation: Condensed liquid.

Q14. If $COP_R = 4$, COP_{HP} is:

(a) 3
(b) 4
(c) 5
(d) 0.25

✔ **Answer: (c) 5**

Explanation: 4 + 1.

Q15. Raising evaporator temperature makes COP:

(a) Decrease
(b) Increase
(c) Constant
(d) Zero

✔ **Answer: (b) Increase**

Explanation: Temperature lift reduces.

SSC JE / UPSSSC JE Previous Year Questions — Principles of Refrigeration Plants

Q1. One ton of refrigeration is equal to:

SSC JE (Previous Year)

(a) 210 kJ/min
(b) 21 kJ/min
(c) 2100 kJ/min
(d) 1 kJ/s

✔ **Answer: (a) 210 kJ/min**

Explanation: ≈ 210–211 kJ/min.

Q2. In VCR, the refrigerant rejects heat in the:

SSC JE (Previous Year)

(a) Evaporator
(b) Condenser
(c) Expansion valve
(d) Compressor

✔ **Answer: (b) Condenser**

Explanation: Condenser.

Q3. Absorbent in the ammonia absorption system: SSC JE (Previous Year)

(a) Ammonia
(b) Water
(c) LiBr
(d) Hydrogen

✔ **Answer: (b) Water**

Explanation: Water.

Q4. Bell–Coleman cycle is a: SSC JE (Previous Year)

(a) Reversed Carnot
(b) Reversed Brayton (Joule)
(c) Otto
(d) Rankine

✔ **Answer: (b) Reversed Brayton (Joule)**

Explanation: Air refrigeration.

Q5. COP of a refrigerator: SSC JE (Previous Year)

(a) Always less than 1
(b) Can be more than 1
(c) Always 1
(d) Negative

✔ **Answer: (b) Can be more than 1**

Explanation: $Q_L > W$ is possible.

Q6. In the expansion valve, constant is: SSC JE (Previous Year)

(a) Entropy
(b) Enthalpy
(c) Pressure
(d) Temperature

✔ **Answer: (b) Enthalpy**

Explanation: Throttling.

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